

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Ticks in boxes 1, 5 and 6 Television 1 uses less energy per second than television 2 ✓ Television 3 uses 40 units more per year than television 4 ✓ Televisions with the same energy rating, e.g A++, don't always have the same power ✓ –1 mark for each additional box ticked		3		3		
	(b)			N.B. Only televisions 1 and 2 to be used. 1st mark – correct substitution of one ratio 2nd mark – correct calculation of one ratio 3rd mark – correct calculation of 2nd ratio 3 marks to be awarded only if correct conclusion present Screen size to screen size compared with power to power $\frac{139}{69} = 2.01$ $\frac{78}{32} = 2.44$ OR $\frac{69}{139} = 0.50$ $\frac{32}{78} = 0.41$ Conclusion – [Ratios not the same] so not true Alternative Ratio of screen size to power compared $\frac{69}{32} = 2.16$ $\frac{139}{78} = 1.78$ OR $\frac{32}{69} = 0.46$ $\frac{78}{139} = 0.56$ Conclusion – [Ratios not the same] so not true Alternative Ratio of screen size to kWh per year compared $\frac{69}{47} = 1.47$ $\frac{139}{108} = 1.29$			3	3	3	

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				OR $\frac{47}{69} = 0.68 \quad \frac{108}{139} = 0.78$ Conclusion – [Ratios not the same] so not true Alternative: Screen size to screen size compared with kWh per year to kWh per year $\frac{69}{139} = 0.50 \quad \frac{47}{108} = 0.44$ OR $\frac{139}{69} = 2.01 \quad \frac{108}{47} = 2.30$ Conclusion – [Ratios not the same] so not true						
	(c)	(i)		Time = $\frac{108}{\left(\frac{78}{1000}\right)}$ (1) substitution [even for $\frac{108}{78}$] Time = 1 384.6 [hours] (1) correct answer correctly rounded Answer = 1.38×10^n where n is not 3 award 1 mark only	1	1		2	2	

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		(ii)		Cost = 108×16 or 108×0.16 (1) substitution Cost = £17.28 (1) answer Accept £17 or £17.00	1	1		2	2	
		(iii)		Running cost of TV 2 for 10 years = $\text{£}17.28 \text{ (ecf)} \times 10 = \text{£}172.80$ (1) Accept £170 or £172 or £173 Running cost of TV 4 for 10 years = $172 \times 10 \times 0.16 = \text{£}275.20$ (1) Accept £275 TV 4 costs £102.40 more to run but it is £200 cheaper to buy so Sarah is right (1) Alternative: Annual savings from using TV 2 = $(172 - 108) \times 0.16 = \text{£}10.24$ (1) Running cost = $\text{£}10.24 \times 10 = \text{£}102.40$ (1) OR Difference in units over 10 years $(172 - 108) \times 10 = 640$ (1) Difference in running cost = $640 \times 0.16 = \text{£}102.40$ (1) 3rd mark - TV 4 costs £102.40 more to run but it is £200 cheaper to buy so Sarah is right (1) Alternative: Total cost of TV 2 = $\text{£}172.80 \text{ ecf (1)} + \text{£}1\,000 = \text{£}1\,172.80$ (1) Total cost of TV 4 = £1 075.20 so cheaper so Sarah is right (1) OR Total cost of TV 4 = $\text{£}275.20 \text{ (1)} + \text{£}800 = \text{£}1\,075.20$ (1) Total cost of TV 2 = £1 172.80 so more expensive so Sarah is right (1) Alternative: Annual savings from using TV 2 = $(172 - 108) \times 0.16 = \text{£}10.24(1)$ Payback time = $\frac{200}{10.24}(1) = 19.5$ years which is longer than 10 years so Sarah is right (1)			3	3	2	

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		(iv)		Fewer power stations / less fuel burnt (1) Reducing CO ₂ emissions / reducing impact on global warming / reduces carbon footprint (1)	2			2		
				Question 1 total	4	5	6	15	9	0